

## A Quantum-Classical Hybrid Approach To Portfolio Optimization

### Introduction & Motivation

As a Computer Science student specializing in Data Analytics, I am driven by the challenge of applying emerging computational methods to quantitative finance. Traditional mean-variance optimization (Markowitz, 1952) faces significant computational hurdles as feature spaces grow. In the Fall of 2025, I sought to address this by exploring Noisy Intermediate-Scale Quantum (NISQ) technologies. Working in a small research group with two graduate students under the mentorship of a doctoral candidate, I investigated whether quantum algorithms could outperform classical benchmarks in identifying high-growth portfolios.

### Methodology and Intellectual Contribution

While our group collaborated on data acquisition, my primary intellectual contribution lay in the mathematical formulation and the hardware implementation of the quantum algorithms.

**Mathematical Formulation:** I was responsible for translating the classical finance problem into a quantum-compatible format. I mathematically derived the formula to map the continuous mean-variance objective onto a Quadratic Unconstrained Binary Optimization (QUBO) model. This required defining a cost function that balanced expected returns against the covariance matrix (risk) using binary variables. To enable execution on the quantum processor, I formulated the energy function as:

$$E(x) = x^T Q x = \sum_i Q_{ii} x_i + \sum_{i < j} Q_{ij} x_i x_j$$

where  $x \in \{0, 1\}^n$  represents the binary decision vector for asset selection, and  $Q \in \mathbb{R}^{n \times n}$  is the upper-triangular matrix encoding the risk-adjusted returns.

**Theoretical Leadership:** To ensure the team understood the underlying mechanics, I mapped out the conceptual framework of the Quantum Approximate Optimization Algorithm (Farhi et al., 2014). I led whiteboarding sessions to derive the specific Ising Hamiltonian required for our circuit, ensuring our cost function was correctly encoded into the phase-separator unitary.

**Hardware Implementation:** Moving from simulation to reality presents significant noise and connectivity challenges. I took the lead in debugging our software stack to ensure compatibility with real quantum hardware. I optimized the circuit depth and resolved transpilation errors, allowing us to successfully execute our algorithm on an actual IBM Quantum processor rather than relying solely on noiseless simulators.

## Results and Performance Metrics

The primary metric for evaluation was the Sharpe Ratio (risk-adjusted return), alongside expected annual return and volatility. Our experiments yielded a critical insight into the current state of NISQ computing: while the quantum algorithm successfully converged to a viable portfolio, the hardware noise introduced significant instability compared to the classical benchmark.

The classical Markowitz model achieved a Sharpe Ratio of approximately 2.37, proving superior in both stability and risk-adjusted efficiency. The QAOA circuit, running on the IBM Quantum processor, identified a portfolio with an expected return (18.45%) nearly identical to the classical baseline (18.63%), but with nearly triple the volatility.

| Method                | Sharpe Ratio | Expected Return | Volatility |
|-----------------------|--------------|-----------------|------------|
| Markowitz (Benchmark) | 2.374        | 18.63%          | 7.85%      |
| QAOA (Quantum)        | 0.827        | 18.45%          | 22.31%     |

Principal Component Analysis (PCA) successfully compressed the feature space, allowing the QAOA circuit to execute within the coherence time of the device. The fact that our quantum algorithm found a portfolio with returns competitive to the classical benchmark (18.45% vs 18.63%) validates the correctness of our Hamiltonian formulation. However, the significantly lower Sharpe ratio (0.827) indicates that gate errors and decoherence prevented the optimizer from fine-tuning the risk parameters as effectively as the classical solver.

## Conclusion and Future Research

This project successfully demonstrated the end-to-end pipeline of financial data ingestion, PCA-based dimensionality reduction, and quantum portfolio optimization.

While the classical Markowitz model maintained the highest stability, the quantum approach (QAOA) successfully demonstrated proof-of-concept viability by converging to a profitable portfolio with returns comparable to the classical ideal. The disparity in volatility (22.31% for Quantum vs 7.85% for Classical) serves as a quantifiable measure of the "noise penalty" currently inherent in NISQ devices.

Having now debugged these circuits on physical hardware, **my proposed future work** focuses on implementing error mitigation techniques. Specifically, I plan to apply Zero-Noise Extrapolation (ZNE) to better estimate the expectation values of the QAOA Hamiltonian. By reducing the noise floor, I aim to improve the Sharpe ratio of the quantum solution, moving it from a theoretical demonstration to a competitively stable financial tool.

## **Bibliography**

Farhi, E., Goldstone, J., & Gutmann, S. (2014). *A quantum approximate optimization algorithm*. arXiv. <https://doi.org/10.48550/arXiv.1411.4028>

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>